Tasks to be Done: **NAME: LOGESHWARI**

ROLL NO: E25AI121

Perform Basic SQL Queries

Concept:

SQL SELECT command is used to retrieve data from database tables.

1. Display all customer details.

```
SQL> SELECT * FROM CUSTOMER;

        ID NAME                                AGE
---------- ------------------------------ ----------
ADDRESS                        SALARY
------------------------------ ----------
         1 Ramesh                               32
Ahmedabad                       30000

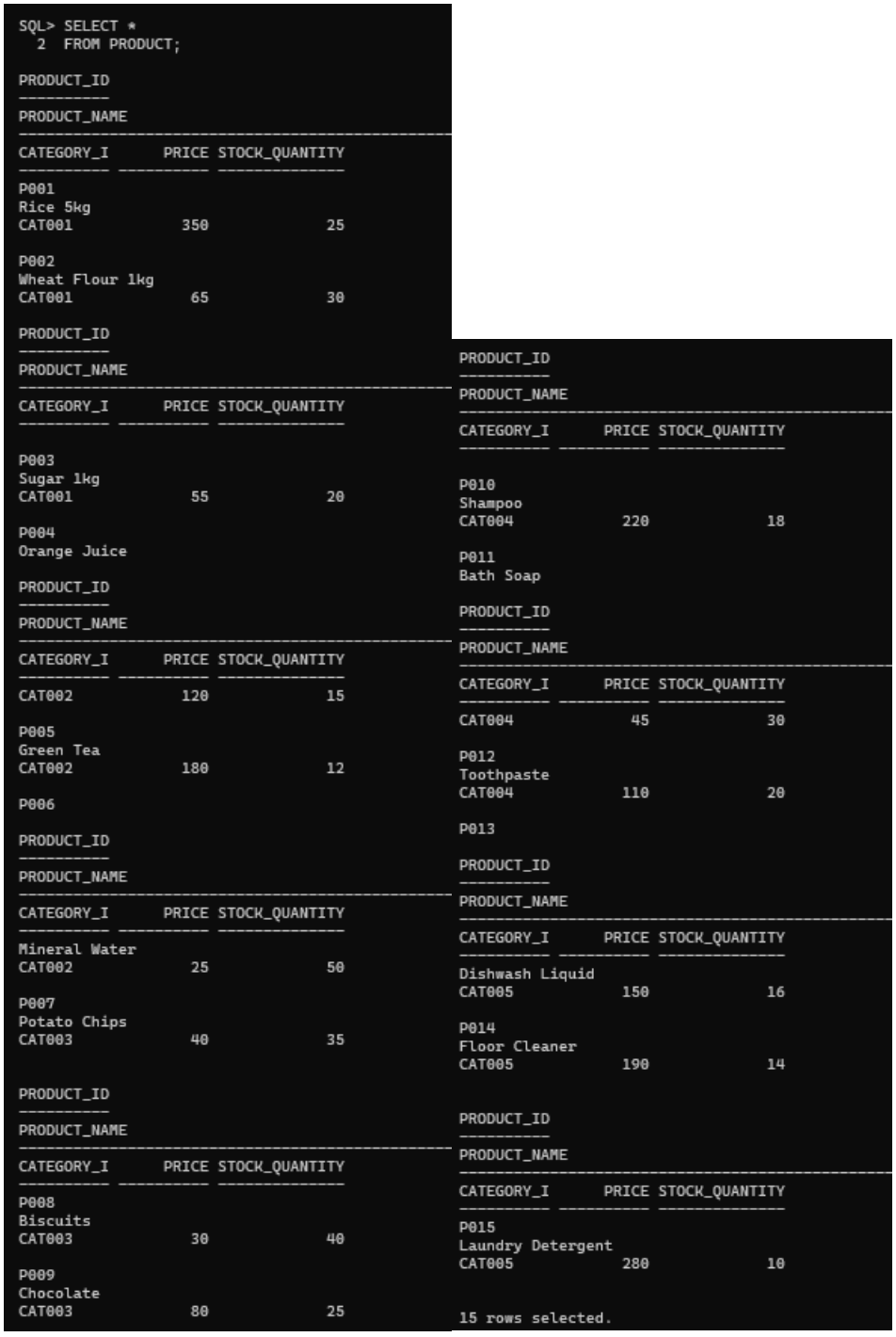
         2 Khilan                               25
Delhi                           30000

         3 kaushik                              23
Kota                            30000


        ID NAME                                AGE
---------- ------------------------------ ----------
ADDRESS                        SALARY
------------------------------ ----------
         4 Chaitali                             25
Mumbai                          75000

         5 Hardik                               27
Bhopal                          80000
```

2.Display all available products.



```
SQL> SELECT *
  2  FROM PRODUCT;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
P001
Rice 5kg
CAT001            350             25

P002
Wheat Flour 1kg
CAT001             65             30

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------

P003
Sugar 1kg
CAT001             55             20

P004
Orange Juice

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
CAT002            120             15

P005
Green Tea
CAT002            180             12

P006

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
Mineral Water
CAT002             25             50

P007
Potato Chips
CAT003             40             35


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
P008
Biscuits
CAT003             30             40

P009
Chocolate
CAT003             80             25

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------

P010
Shampoo
CAT004            220             18

P011
Bath Soap

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
CAT004             45             30

P012
Toothpaste
CAT004            110             20

P013

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
Dishwash Liquid
CAT005            150             16

P014
Floor Cleaner
CAT005            190             14


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
P015
Laundry Detergent
CAT005            280             10


15 rows selected.
```

3.Retrieve product names and prices only.

```
SQL> SELECT PRODUCT_NAME, PRICE
  2  FROM PRODUCT;

PRODUCT_NAME
------------------------------------
     PRICE
----------
Rice 5kg
       350

Wheat Flour 1kg
        65

Sugar 1kg
        55


PRODUCT_NAME
------------------------------------
     PRICE
----------
Orange Juice
       120

Green Tea
       180

Mineral Water
        25


PRODUCT_NAME
------------------------------------
     PRICE
----------
Potato Chips
        40

Biscuits
        30

Chocolate
        80


PRODUCT_NAME
------------------------------------
     PRICE
----------
Shampoo
       220

Bath Soap
        45

Toothpaste
       110


PRODUCT_NAME
------------------------------------
     PRICE
----------
Dishwash Liquid
       150

Floor Cleaner
       190

Laundry Detergent
       280


15 rows selected.
```

4. Display all orders placed by customers.

```
SQL> SELECT *
  2  FROM SALES;

SALE_ID    CUSTOMER_I SALE_DATE TOTAL_AMOUNT PAYMENT_STATUS
---------- ---------- --------- ------------ ----------------
S001       C001       01-SEP-26          415 Paid
S002       C002       02-SEP-26          300 Paid
S003       C003       03-SEP-26          340 Paid
S004       C004       04-SEP-26          470 Paid
S005       C005       05-SEP-26          360 Paid
```

5. Retrieve payment details.

```
SQL> SELECT *
  2  FROM PAYMENT;

PAYMENT_ID SALE_ID    PAYMENT_D PAYMENT_MODE       PAYMENT_STATUS
---------- ---------- --------- ------------------ ----------------
TRANSACTION_AMOUNT
------------------
PAY001     S001       01-SEP-26 UPI                SUCCESS
               415

PAY002     S002       02-SEP-26 CASH               SUCCESS
               300

PAY003     S003       03-SEP-26 CARD               SUCCESS
               340


PAYMENT_ID SALE_ID    PAYMENT_D PAYMENT_MODE       PAYMENT_STATUS
---------- ---------- --------- ------------------ ----------------
TRANSACTION_AMOUNT
------------------
PAY004     S004       04-SEP-26 UPI                SUCCESS
               470

PAY005     S005       05-SEP-26 CASH               SUCCESS
               360
```

2. Apply Filtering Conditions Using WHERE

1. Find products with price greater than ₹5000.

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE PRICE > 5000;

no rows selected
```

2.Display products available in stock.

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE STOCK_QUANTITY > 0;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------

P001
Rice 5kg
CAT001            350             25

P002
Wheat Flour 1kg
CAT001             65             30

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------

P003
Sugar 1kg
CAT001             55             20

P004
Orange Juice

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
CAT002            120             15

P005
Green Tea
CAT002            180             12

P006

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
Mineral Water
CAT002             25             50

P007
Potato Chips
CAT003             40             35
```

```
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------

P008
Biscuits
CAT003             30             40

P009
Chocolate
CAT003             80             25

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------

P010
Shampoo
CAT004            220             18

P011
Bath Soap

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
CAT004             45             30

P012
Toothpaste
CAT004            110             20

P013

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
Dishwash Liquid
CAT005            150             16

P014
Floor Cleaner
CAT005            190             14


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
P015
Laundry Detergent
CAT005            280             10


15 rows selected.
```

3.Display completed orders.

```
SQL> SELECT *
  2  FROM SALES
  3  WHERE PAYMENT_STATUS = 'Completed';

no rows selected
```

4.Find products with rating above 4.

```
SQL> SELECT *
  2  FROM PRODUCT_REVIEW
  3  WHERE RATING > 4;

 REVIEW_ID CUSTOMER_ID PRODUCT_ID     RATING
---------- ----------- ---------- ----------
REVIEW_COMMENT
--------------------------------------------------
REVIEW_DA
---------
         1           1 P001                5
Excellent product
01-NOV-26

         4           4 P004                5
Very refreshing juice
01-NOV-26

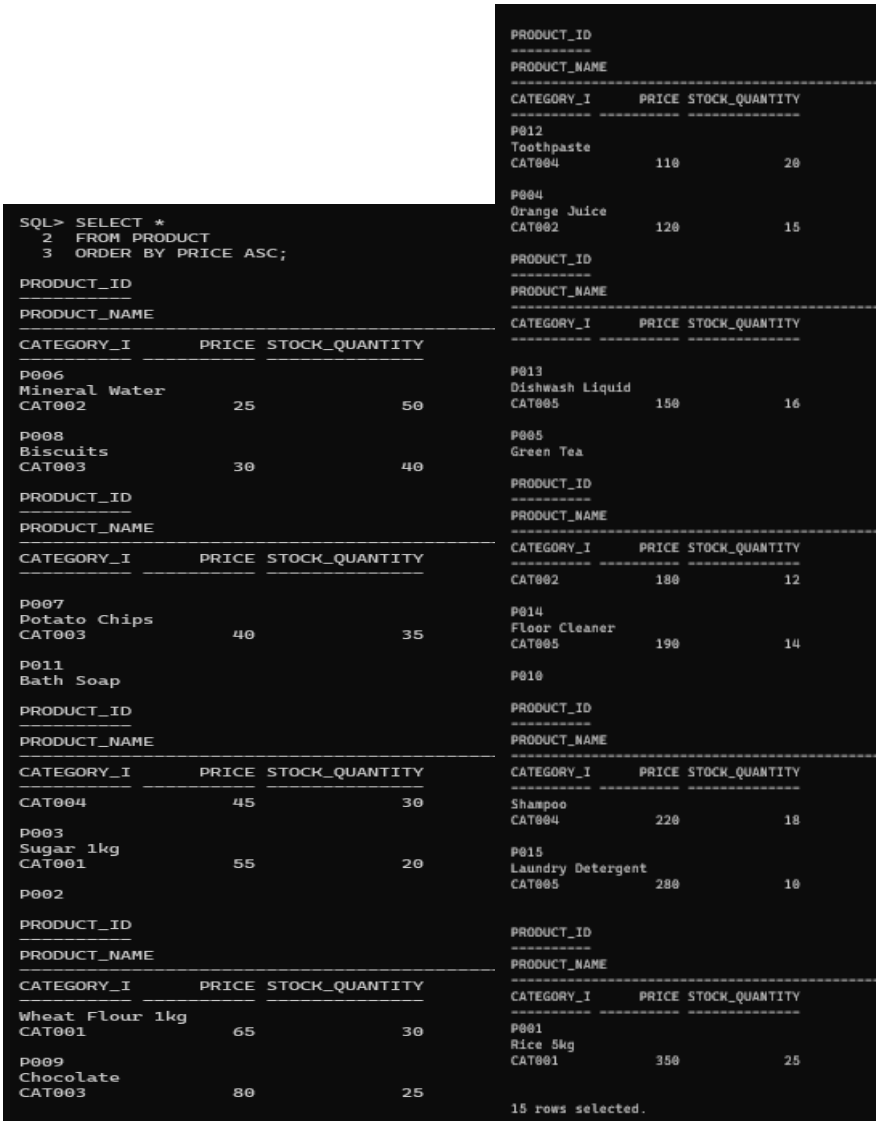
 REVIEW_ID CUSTOMER_ID PRODUCT_ID     RATING
---------- ----------- ---------- ----------
REVIEW_COMMENT
--------------------------------------------------
REVIEW_DA
---------

         7           1 P001                5
Excellent product
01-NOV-26
```

3. Sort Data Using ORDER BY

Tasks:

1. Display products from lowest to highest price.



```
SQL> SELECT *
  2  FROM PRODUCT
  3  ORDER BY PRICE ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
P006
Mineral Water
CAT002             25             50

P008
Biscuits
CAT003             30             40

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------

P007
Potato Chips
CAT003             40             35

P011
Bath Soap

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
CAT004             45             30

P003
Sugar 1kg
CAT001             55             20

P002

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
Wheat Flour 1kg
CAT001             65             30

P009
Chocolate
CAT003             80             25

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
P012
Toothpaste
CAT004            110             20

P004
Orange Juice
CAT002            120             15

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------

P013
Dishwash Liquid
CAT005            150             16

P005
Green Tea

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
CAT002            180             12

P014
Floor Cleaner
CAT005            190             14

P010

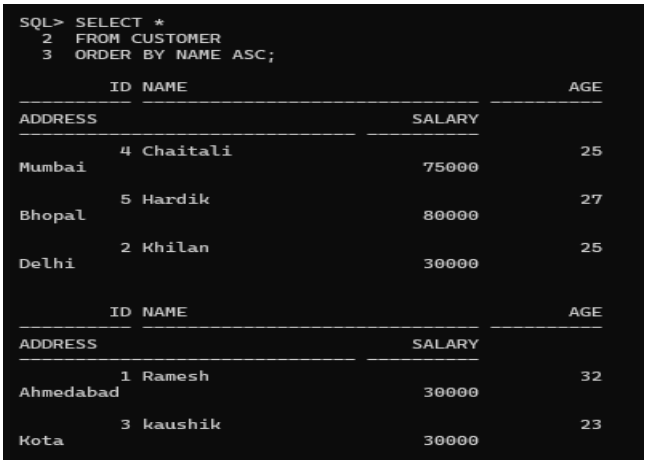
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
Shampoo
CAT004            220             18

P015
Laundry Detergent
CAT005            280             10


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
P001
Rice 5kg
CAT001            350             25

15 rows selected.
```

2. Display customers alphabetically.



```
SQL> SELECT *
  2  FROM CUSTOMER
  3  ORDER BY NAME ASC;

        ID NAME                                    AGE
---------- ------------------------------ ----------
ADDRESS                        SALARY
------------------------------ ----------
         4 Chaitali                                25
Mumbai                          75000

         5 Hardik                                  27
Bhopal                          80000

         2 Khilan                                  25
Delhi                           30000


        ID NAME                                    AGE
---------- ------------------------------ ----------
ADDRESS                        SALARY
------------------------------ ----------
         1 Ramesh                                  32
Ahmedabad                       30000

         3 kaushik                                 23
Kota                            30000
```

3. Find top expensive products.

```
SQL> SELECT *
  2  FROM (
  3      SELECT *
  4      FROM PRODUCT
  5      ORDER BY PRICE DESC
  6  )
  7  WHERE ROWNUM <= 5;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
P001
Rice 5kg
CAT001            350             25

P015
Laundry Detergent
CAT005            280             10

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------


P010
Shampoo
CAT004            220             18

P014
Floor Cleaner

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
CAT005            190             14

P005
Green Tea
CAT002            180             12
```

4. Display latest orders first.

```
SQL> SELECT *
  2  FROM SALES
  3  ORDER BY SALE_DATE DESC;

SALE_ID    CUSTOMER_I SALE_DATE TOTAL_AMOUNT PAYMENT_STATUS
---------- ---------- --------- ------------ ----------------
S005       C005       05-SEP-26          360 Paid
S004       C004       04-SEP-26          470 Paid
S003       C003       03-SEP-26          340 Paid
S002       C002       02-SEP-26          300 Paid
S001       C001       01-SEP-26          415 Paid
```

4. Retrieve Unique Values Using DISTINCT

Tasks:

1.Display unique product categories.

```
SQL> SELECT DISTINCT CATEGORY_ID
  2  FROM PRODUCT;

CATEGORY_I
----------
CAT003
CAT002
CAT004
CAT001
CAT005
```

2. Find different payment methods used by customers.

```
SQL> SELECT DISTINCT PAYMENT_MODE
  2  FROM PAYMENT;

PAYMENT_MODE
--------------------
CASH
CARD
UPI
```

5. Search Products Based on Conditions

Tasks:

Write SQL queries to:

1. Find products within a specific price range. (Product Between 20,000 to 40,000)

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE PRICE BETWEEN 20000 AND 40000;

no rows selected
```

2.Display products belonging to a particular category.

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE CATEGORY_ID = 'C001';

no rows selected
```

3.Find products currently available in inventory.

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE STOCK_QUANTITY > 0;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
P001
Rice 5kg
CAT001            350             25

P002
Wheat Flour 1kg
CAT001             65             30

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------

P003
Sugar 1kg
CAT001             55             20

P004
Orange Juice

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
CAT002            120             15

P005
Green Tea
CAT002            180             12

P006

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
Mineral Water
CAT002             25             50

P007
Potato Chips
CAT003             40             35


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
P008
Biscuits
CAT003             30             40

P009
Chocolate
CAT003             80             25

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------

P010
Shampoo
CAT004            220             18

P011
Bath Soap

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
CAT004             45             30

P012
Toothpaste
CAT004            110             20

P013

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
Dishwash Liquid
CAT005            150             16

P014
Floor Cleaner
CAT005            190             14


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_I      PRICE STOCK_QUANTITY
---------- ---------- --------------
P015
Laundry Detergent
CAT005            280             10


15 rows selected.
```

4.Search products using product name.

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE PRODUCT_NAME LIKE '%Mobile%';

no rows selected
```

5.Find low-stock products.

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE STOCK_QUANTITY < 10;

no rows selected
```

6. Retrieve Customer and Product Information

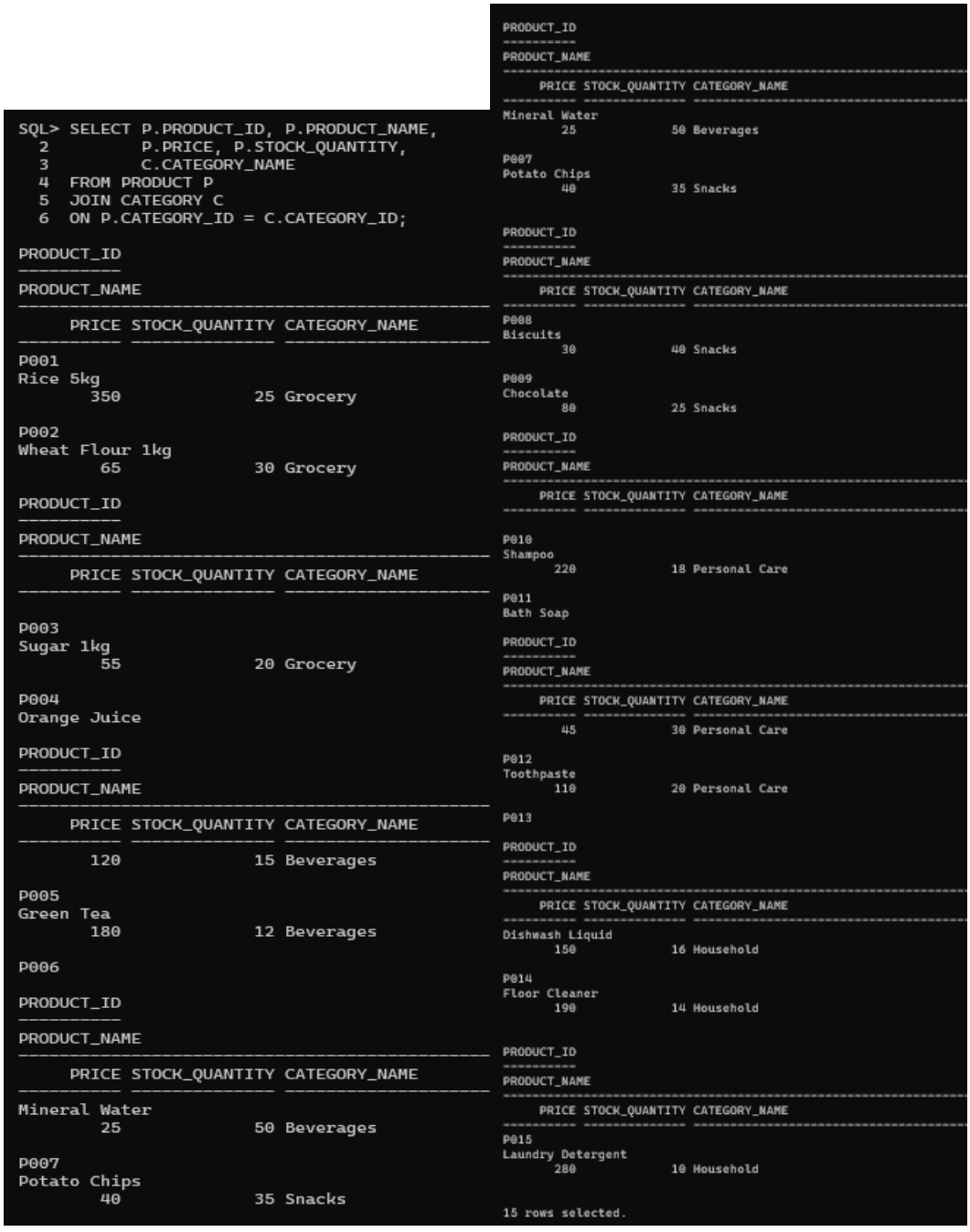
Tasks:

1. Display customer details with their orders.

```
SQL> SELECT C.ID, C.NAME, S.SALE_ID,
  2         S.SALE_DATE, S.TOTAL_AMOUNT,
  3         S.PAYMENT_STATUS
  4  FROM CUSTOMER C
  5  JOIN SALES S
  6  ON TO_CHAR(C.ID) = S.CUSTOMER_ID;

no rows selected
```

2.Display product details with category information.



```
SQL> SELECT P.PRODUCT_ID, P.PRODUCT_NAME,
  2         P.PRICE, P.STOCK_QUANTITY,
  3         C.CATEGORY_NAME
  4  FROM PRODUCT P
  5  JOIN CATEGORY C
  6  ON P.CATEGORY_ID = C.CATEGORY_ID;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE STOCK_QUANTITY CATEGORY_NAME
---------- -------------- --------------------
P001
Rice 5kg
       350             25 Grocery

P002
Wheat Flour 1kg
        65             30 Grocery

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE STOCK_QUANTITY CATEGORY_NAME
---------- -------------- --------------------

P003
Sugar 1kg
        55             20 Grocery

P004
Orange Juice

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE STOCK_QUANTITY CATEGORY_NAME
---------- -------------- --------------------
       120             15 Beverages

P005
Green Tea
       180             12 Beverages

P006

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE STOCK_QUANTITY CATEGORY_NAME
---------- -------------- --------------------
Mineral Water
        25             50 Beverages

P007
Potato Chips
        40             35 Snacks

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE STOCK_QUANTITY CATEGORY_NAME
---------- -------------- --------------------
Mineral Water
        25             50 Beverages

P007
Potato Chips
        40             35 Snacks


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE STOCK_QUANTITY CATEGORY_NAME
---------- -------------- --------------------
P008
Biscuits
        30             40 Snacks

P009
Chocolate
        80             25 Snacks

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE STOCK_QUANTITY CATEGORY_NAME
---------- -------------- --------------------

P010
Shampoo
       220             18 Personal Care

P011
Bath Soap

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE STOCK_QUANTITY CATEGORY_NAME
---------- -------------- --------------------
        45             30 Personal Care

P012
Toothpaste
       110             20 Personal Care

P013

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE STOCK_QUANTITY CATEGORY_NAME
---------- -------------- --------------------
Dishwash Liquid
       150             16 Household

P014
Floor Cleaner
       190             14 Household


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE STOCK_QUANTITY CATEGORY_NAME
---------- -------------- --------------------
P015
Laundry Detergent
       280             10 Household

15 rows selected.
```

3. Find customers who purchased a specific product.

```
SQL> SELECT DISTINCT C.ID, C.NAME
  2  FROM CUSTOMER C
  3  JOIN SALES S
  4  ON TO_CHAR(C.ID) = S.CUSTOMER_ID
  5  WHERE S.SALE_ID IN (
  6      SELECT SALE_ID
  7      FROM SALES
  8  )
  9  ;

no rows selected

SQL>
```

4.Display Products Purchased by Each Customer

```
SQL> SELECT C.ID, C.NAME,
  2         P.PRODUCT_ID, P.PRODUCT_NAME,
  3         SD.QUANTITY, SD.PRICE
  4  FROM CUSTOMER C
  5  JOIN SALES S
  6  ON 'C' || LPAD(TO_CHAR(C.ID), 3, '0') = S.CUSTOMER_ID
  7  JOIN SALE_DETAILS SD
  8  ON S.SALE_ID = SD.SALE_ID
  9  JOIN PRODUCT P
 10  ON SD.PRODUCT_ID = P.PRODUCT_ID
 11  ORDER BY C.NAME;

        ID NAME                           PRODUCT_ID
---------- ------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
         4 Chaitali                       P010
Shampoo
         1        220

         4 Chaitali                       P009
Chocolate
         1         80

        ID NAME                           PRODUCT_ID
---------- ------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------

         4 Chaitali                       P005
Green Tea
         1        180

         5 Hardik                         P006
Mineral Water

        ID NAME                           PRODUCT_ID
---------- ------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
         1         25

         5 Hardik                         P008
Biscuits
         1         30

         5 Hardik                         P015
```

```
        ID NAME                           PRODUCT_ID
---------- ------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
Laundry Detergent
         1        280

         2 Khilan                         P005
Green Tea
         1        180


        ID NAME                           PRODUCT_ID
---------- ------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
         2 Khilan                         P004
Orange Juice
         1        120

         1 Ramesh                         P002
Wheat Flour 1kg
         1         65

        ID NAME                           PRODUCT_ID
---------- ------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------

         1 Ramesh                         P001
Rice 5kg
         1        350

         3 kaushik                        P007
Potato Chips

        ID NAME                           PRODUCT_ID
---------- ------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
         1         40

         3 kaushik                        P008
Biscuits
         1         30

         3 kaushik                        P003

        ID NAME                           PRODUCT_ID
---------- ------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
Sugar 1kg
         2         55


13 rows selected.
```

7. Apply Multiple Filtering Conditions

Tasks:

1. Find electronics products costing more than ₹10000.

```
SQL> SELECT *
  2  FROM PRODUCT P
  3  JOIN CATEGORY C
  4  ON P.CATEGORY_ID = C.CATEGORY_ID
  5  WHERE C.CATEGORY_NAME = 'Electronics'
  6  AND P.PRICE > 10000;

no rows selected
```

2.Search Products Containing the Word "Mobile"

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE PRODUCT_NAME LIKE '%Mobile%';

no rows selected
```

2. Find Orders Placed Within a Specific Date Range

```
SQL> SELECT
  2      SALE_ID,
  3      CUSTOMER_ID,
  4      SALE_DATE,
  5      TOTAL_AMOUNT,
  6      PAYMENT_STATUS
  7  FROM SALES
  8  WHERE SALE_DATE BETWEEN
  9      TO_DATE('01-SEP-2026', 'DD-MON-YYYY')
 10      AND TO_DATE('30-SEP-2026', 'DD-MON-YYYY')
 11  ORDER BY SALE_DATE;

SALE_ID    CUSTOMER_I SALE_DATE TOTAL_AMOUNT PAYMENT_STATUS
---------- ---------- --------- ------------ --------------
S001       C001       01-SEP-26          415 Paid
S002       C002       02-SEP-26          300 Paid
S003       C003       03-SEP-26          340 Paid
S004       C004       04-SEP-26          470 Paid
S005       C005       05-SEP-26          360 Paid
```

8. Generate Basic Business Reports

Report 1: Product Availability Report

Display:

• Product name

• Price

• Stock quantity

• Availability status

```
SQL> SELECT
  2      PRODUCT_NAME,
  3      PRICE,
  4      STOCK_QUANTITY,
  5      CASE
  6          WHEN STOCK_QUANTITY > 0 THEN 'Available'
  7          ELSE 'Out of Stock'
  8      END AS AVAILABILITY_STATUS
  9  FROM PRODUCT
 10  ORDER BY PRODUCT_NAME;

PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- -------------
Bath Soap
        45             30 Available

Biscuits
        30             40 Available

Chocolate
        80             25 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- -------------
Dishwash Liquid
       150             16 Available

Floor Cleaner
       190             14 Available

Green Tea
       180             12 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- -------------
Laundry Detergent
       280             10 Available

Mineral Water
        25             50 Available

Orange Juice
       120             15 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- -------------
Potato Chips
        40             35 Available

Rice 5kg
       350             25 Available

Shampoo
       220             18 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- -------------
Sugar 1kg
        55             20 Available

Toothpaste
       110             20 Available

Wheat Flour 1kg
        65             30 Available


15 rows selected.
```

Report 2: Customer Report

Generate:

● Total number of customers

```
SQL> SELECT
  2      COUNT(*) AS TOTAL_NUMBER_OF_CUSTOMERS
  3  FROM CUSTOMER;

TOTAL_NUMBER_OF_CUSTOMERS
-------------------------
                        5
```

● New customer registrations

```
SQL> SELECT
  2      ID,
  3      NAME,
  4      AGE,
  5      ADDRESS,
  6      SALARY
  7  FROM CUSTOMER
  8  ORDER BY ID DESC;

        ID NAME                                AGE
---------- ------------------------------ ----------
ADDRESS                         SALARY
------------------------------ ----------
         5 Hardik                               27
Bhopal                           80000

         4 Chaitali                             25
Mumbai                           75000

         3 kaushik                              23
Kota                             30000


        ID NAME                                AGE
---------- ------------------------------ ----------
ADDRESS                         SALARY
------------------------------ ----------
         2 Khilan                               25
Delhi                            30000

         1 Ramesh                               32
Ahmedabad                        30000
```

Report 3: Order Report

Display:

● Total orders

```
SQL> SELECT
  2      COUNT(*) AS TOTAL_NUMBER_OF_ORDERS
  3  FROM SALES;

TOTAL_NUMBER_OF_ORDERS
----------------------
                     5
```

● Completed orders

```
SQL> SELECT
  2      COUNT(*) AS COMPLETED_ORDERS
  3  FROM SALES
  4  WHERE PAYMENT_STATUS = 'Completed';

COMPLETED_ORDERS
----------------
               0
```

● Pending orders

```
SQL> SELECT
  2      COUNT(*) AS PENDING_ORDERS
  3  FROM SALES
  4  WHERE PAYMENT_STATUS = 'Pending';

PENDING_ORDERS
--------------
             0
```

● Cancelled orders

```
SQL> SELECT
  2      COUNT(*) AS CANCELLED_ORDERS
  3  FROM SALES
  4  WHERE PAYMENT_STATUS = 'Cancelled';

CANCELLED_ORDERS
----------------
               0
```

Report 4: Product Performance Report

Find:

● Highest-priced products

```
SQL> SELECT
  2      PRODUCT_ID,
  3      PRODUCT_NAME,
  4      PRICE
  5  FROM PRODUCT
  6  ORDER BY PRICE DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------
     PRICE
----------
P001
Rice 5kg
       350

P015
Laundry Detergent
       280

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------
     PRICE
----------

P010
Shampoo
       220

P014
Floor Cleaner

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------
     PRICE
----------
       190

P005
Green Tea
       180

P013

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------
     PRICE
----------
Dishwash Liquid
       150

P004
Orange Juice
       120

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------
     PRICE
----------
P012
Toothpaste
       110

P009
Chocolate
        80

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------
     PRICE
----------

P002
Wheat Flour 1kg
        65

P003
Sugar 1kg

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------
     PRICE
----------
        55

P011
Bath Soap
        45

P007

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------
     PRICE
----------
Potato Chips
        40

P008
Biscuits
        30


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------
     PRICE
----------
P006
Mineral Water
        25


15 rows selected.
```

● Most reviewed products

```
SQL> SELECT
  2      P.PRODUCT_NAME,
  3      COUNT(PR.REVIEW_ID) AS TOTAL_REVIEWS
  4  FROM PRODUCT P
  5  JOIN PRODUCT_REVIEW PR
  6  ON P.PRODUCT_ID = PR.PRODUCT_ID
  7  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  8  ORDER BY TOTAL_REVIEWS DESC;

PRODUCT_NAME
--------------------------------------------------
TOTAL_REVIEWS
-------------
Rice 5kg
            3

Sugar 1kg
            1

Orange Juice
            1


PRODUCT_NAME
--------------------------------------------------
TOTAL_REVIEWS
-------------
Green Tea
            1

Wheat Flour 1kg
            1
```

● Available products

```
SQL> SELECT
  2      PRODUCT_ID,
  3      PRODUCT_NAME,
  4      PRICE,
  5      STOCK_QUANTITY
  6  FROM PRODUCT
  7  WHERE STOCK_QUANTITY > 0
  8  ORDER BY PRODUCT_NAME;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P011
Bath Soap
        45             30

P008
Biscuits
        30             40

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------
     PRICE STOCK_QUANTITY
---------- --------------

P009
Chocolate
        80             25

P013
Dishwash Liquid

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
       150             16

P014
Floor Cleaner
       190             14

P005

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
Green Tea
       180             12

P015
Laundry Detergent
       280             10

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P006
Mineral Water
        25             50

P004
Orange Juice
       120             15

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------
     PRICE STOCK_QUANTITY
---------- --------------

P007
Potato Chips
        40             35

P001
Rice 5kg

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
       350             25

P010
Shampoo
       220             18

P003

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
Sugar 1kg
        55             20

P012
Toothpaste
       110             20


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P002
Wheat Flour 1kg
        65             30


15 rows selected.
```